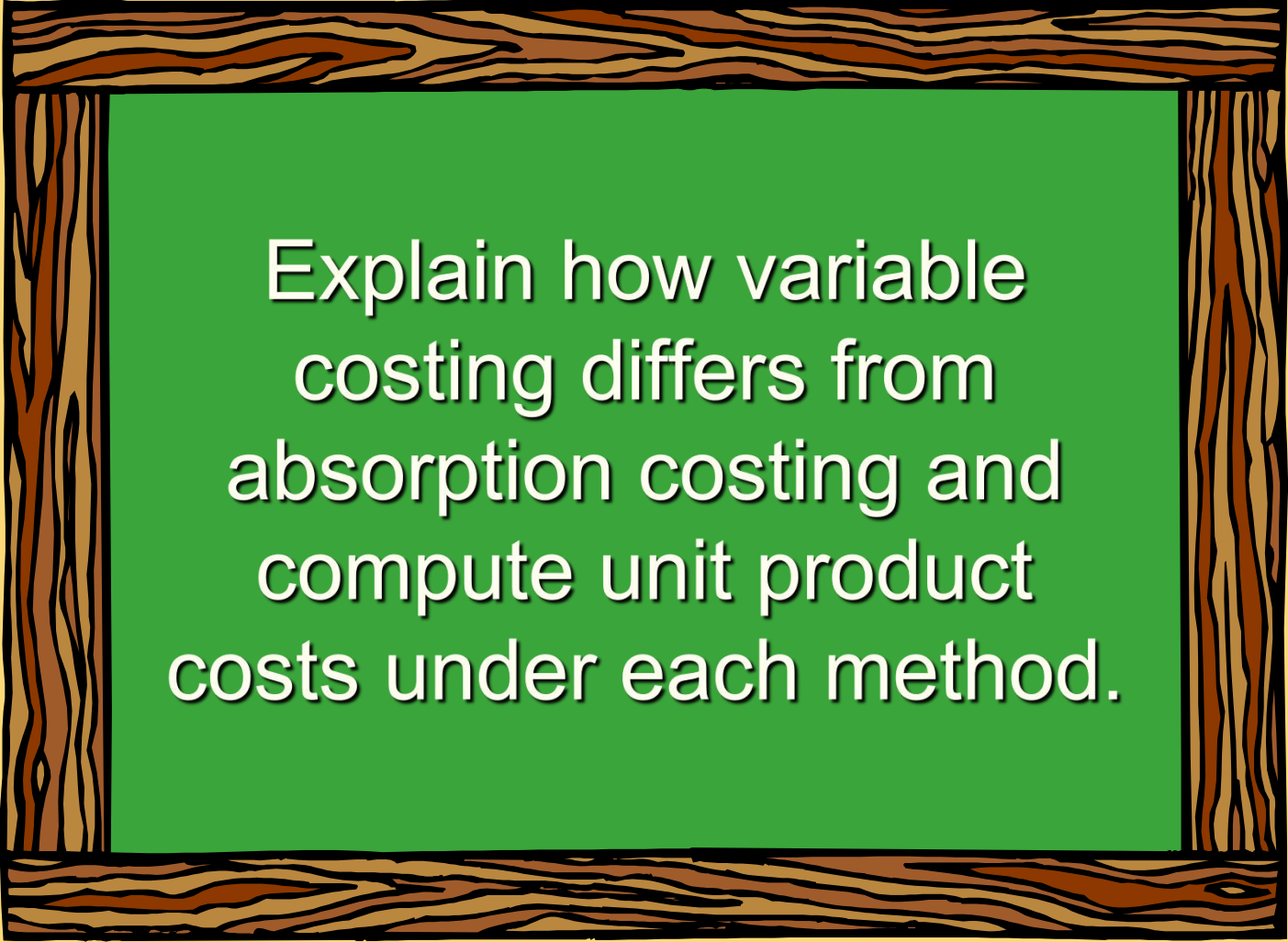


Variable Costing: A Tool for Management

Chapter 7



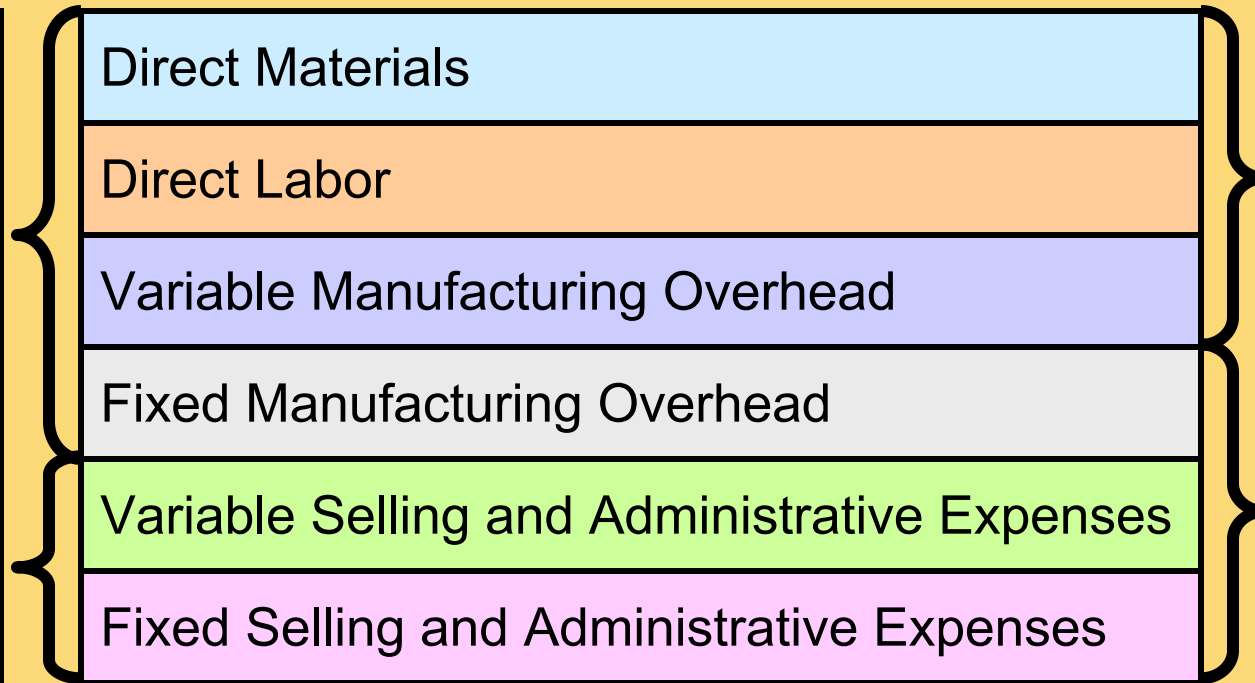
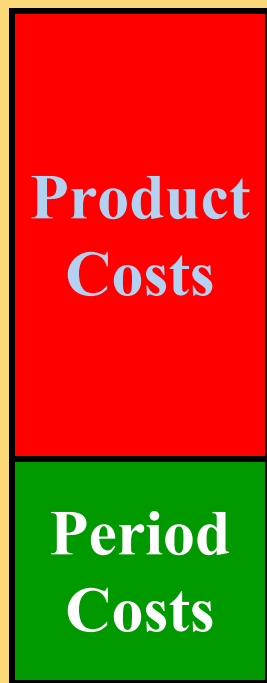
Learning Objective 1



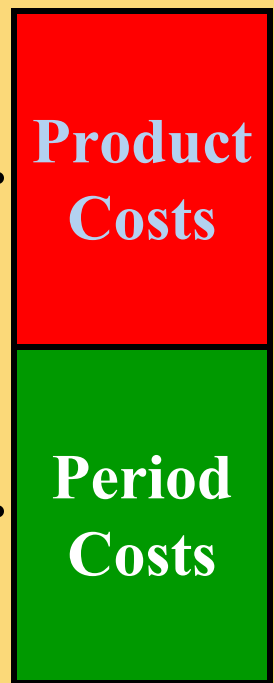
Explain how variable costing differs from absorption costing and compute unit product costs under each method.

Overview of Absorption and Variable Costing

Absorption Costing



Variable Costing



Quick Check ✓

Which method will produce the highest values for work in process and finished goods inventories?

- a. Absorption costing.
- b. Variable costing.
- c. They produce the same values for these inventories.
- d. It depends. . .

Quick Check ✓

Which method will produce the highest values for work in process and finished goods inventories?

- a. Absorption costing.
- b. Variable costing.
- c. They produce the same values for these inventories.
- d. It depends. . .

Unit Cost Computations

Harvey Company produces a single product with the following information available:

Number of units produced annually	25,000
--	---------------

Variable costs per unit:

Direct materials, direct labor, and variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3

Fixed costs per year:

Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

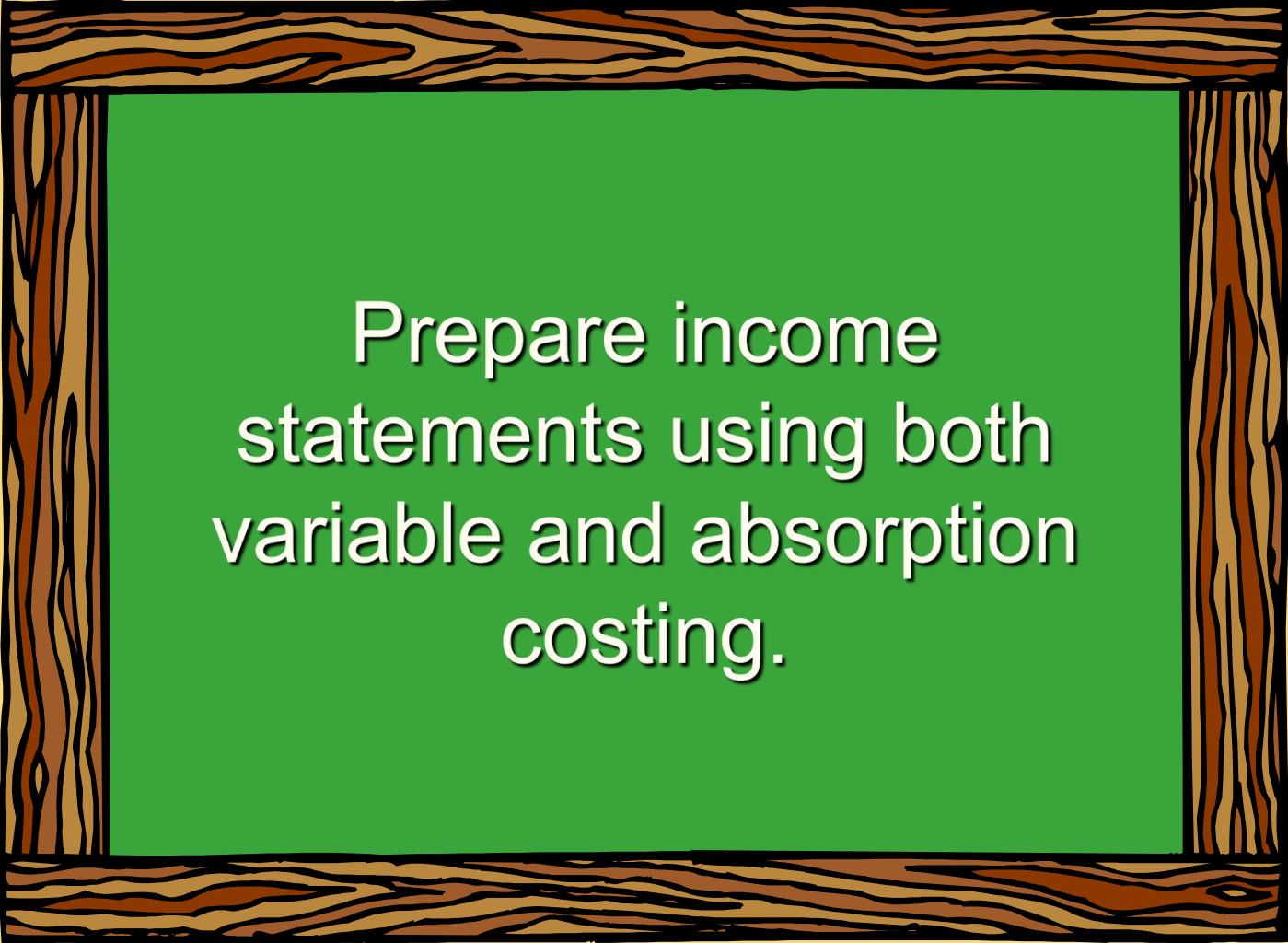
Unit Cost Computations

Unit **product cost** is determined as follows:

	Absorption Costing	Variable Costing
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead (\$150,000 ÷ 25,000 units)	6	-
Unit product cost	<u>\$ 16</u>	<u>\$ 10</u>

Under *absorption costing*, selling and administrative expenses are always treated as **period expenses** and deducted from revenue as incurred.

Learning Objective 2



Prepare income
statements using both
variable and absorption
costing.

Income Comparison of Absorption and Variable Costing

Let's assume the following additional information for Harvey Company.

- ◆ 20,000 units were sold during the year at a price of \$30 each.
- ◆ There were no units in beginning inventory.

Now, let's compute net operating income using both absorption and variable costing.



Absorption Costing

	<u>Absorption Costing</u>	
Sales (20,000 × \$30)		\$ 600,000
Less cost of goods sold:		
Beginning inventory	\$ -	
Add COGM (25,000 × \$16)	400,000	
Goods available for sale	<u>400,000</u>	
Ending inventory (5,000 × \$16)	80,000	320,000
Gross margin		<u>280,000</u>
Less selling & admin. exp.		
Variable (20,000 × \$3)	\$ 60,000	
Fixed	100,000	160,000
Net operating income		<u><u>\$ 120,000</u></u>

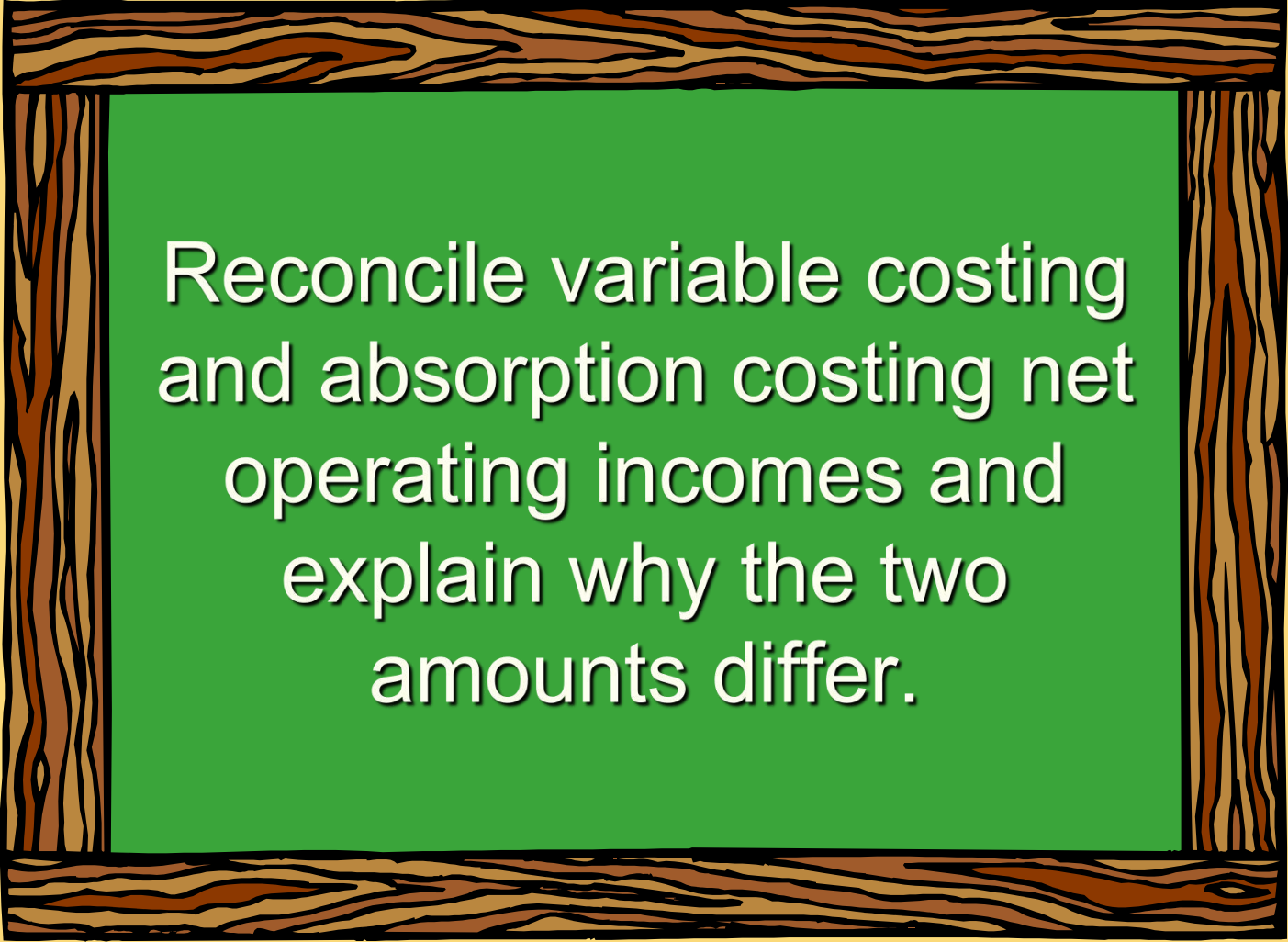
Variable Costing

		Variable Costing	
		<u>\$ 600,000</u>	
Sales (20,000 × \$30)			
Less variable expenses:			
Beginning inventory	\$ -		
Add COGM (25,000 × \$10)	<u>250,000</u>		
Goods available for sale	250,000		
Less ending inventory (5,000 × \$10)	<u>50,000</u>		
Variable cost of goods sold	200,000		
Variable selling & administrative expenses (20,000 × \$3)	<u>60,000</u>		
Contribution margin		<u>260,000</u>	
Less fixed expenses:			<u>340,000</u>
Manufacturing overhead	\$ 150,000		
Selling & administrative expenses	<u>100,000</u>		<u>250,000</u>
Net operating income			<u><u>\$ 90,000</u></u>

**Variable
manufacturing
costs only.**

**All fixed
manufacturing
overhead is
expensed.**

Learning Objective 3



Reconcile variable costing and absorption costing net operating incomes and explain why the two amounts differ.

Comparing the Two Methods

	<u>Cost of Goods Sold</u>	<u>Ending Inventory</u>	<u>Period Expense</u>	<u>Total</u>
Absorption costing				
Variable mfg. costs	\$ 200,000	\$ 50,000	\$ -	\$ 250,000
Fixed mfg. costs	120,000	30,000	-	150,000
	<u>\$ 320,000</u>	<u>\$ 80,000</u>	<u>\$ -</u>	<u>\$ 400,000</u>
Variable costing				
Variable mfg. costs	\$ 200,000	\$ 50,000	\$ -	\$ 250,000
Fixed mfg. costs	-	-	150,000	150,000
	<u>\$ 200,000</u>	<u>\$ 50,000</u>	<u>\$ 150,000</u>	<u>\$ 400,000</u>

Comparing the Two Methods

We can reconcile the difference between absorption and variable income as follows:

Variable costing net operating income	\$ 90,000
Add: Fixed mfg. overhead costs deferred in inventory (5,000 units × \$6 per unit)	30,000
Absorption costing net operating income	<u>\$ 120,000</u>

$$\frac{\text{Fixed mfg. Overhead}}{\text{Units produced}} = \frac{\$150,000}{25,000 \text{ units}} = \$6.00 \text{ per unit}$$

Extended Comparisons of Income Data Harvey Company Year Two

Number of units produced	25,000
Number of units sold	30,000
Units in beginning inventory	5,000
Unit sales price	\$ 30

Variable costs per unit:

Direct materials, direct labor variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3

Fixed costs per year:

Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

Unit Cost Computations

	Absorption Costing	Variable Costing
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead (\$150,000 ÷ 25,000 units)	6	-
Unit product cost	\$ 16	\$ 10

Since there was no change in the variable costs per unit, total fixed costs, or the number of units produced, the unit costs remain unchanged.

Absorption Costing

	Absorption Costing	
Sales (30,000 × \$30)		\$ 900,000
Less cost of goods sold:		
Beg. inventory (5,000 × \$16)	\$ 80,000	
Add COGM (25,000 × \$16)	400,000	
Goods available for sale	480,000	
Less ending inventory	-	480,000
Gross margin		420,000
Less selling & admin. exp.		
Variable (30,000 × \$3)	\$ 90,000	
Fixed	100,000	190,000
Net operating income		\$ 230,000

These are the 25,000 units produced in the current period.

Variable Costing

**Variable
manufacturing
costs only.**

Variable Costing

Sales (30,000 × \$30)		\$ 900,000
Less variable expenses:		
Beg. inventory (5,000 × \$10)	\$ 50,000	
Add COGM (25,000 × \$10)	250,000	
Goods available for sale	<u>300,000</u>	
Less ending inventory	-	
Variable cost of goods sold	<u>300,000</u>	
Variable selling & administrative expenses (30,000 × \$3)	90,000	
Contribution margin	<u>510,000</u>	
Less fixed expenses:		
Manufacturing overhead	\$ 150,000	
Selling & administrative expenses	<u>100,000</u>	<u>250,000</u>
Net operating income		<u><u>\$ 260,000</u></u>

**All fixed
manufacturing
overhead is
expensed.**

Comparing the Two Methods

We can reconcile the difference between absorption and variable income as follows:

Variable costing net operating income \$ 260,000

Deduct: Fixed manufacturing overhead
costs released from inventory

(5,000 units × \$6 per unit)

30,000

Absorption costing net operating income \$ 230,000

$$\frac{\text{Fixed mfg. Overhead}}{\text{Units produced}} = \frac{\$150,000}{25,000 \text{ units}} = \$6.00 \text{ per unit}$$

Comparing the Two Methods

Costing Method	1st Period	2nd Period	Total
Absorption	\$ 120,000	\$ 230,000	\$ 350,000
Variable	90,000	260,000	350,000



Summary of Key Insights

Relation between production and sales	Effect on inventory	Relation between variable and absorption income
Production > Sales	Inventory increases	Absorption > Variable
Production < Sales	Inventory decreases	Absorption < Variable
Production = Sales	No change	Absorption = Variable

Effect of Changes in Production on Net Operating Income

Let's revise the Harvey Company example.

In the previous example, 25,000 units were produced each year, but sales increased from 20,000 units in year one to 30,000 units in year two.

In this revised example, production will differ each year while sales will remain constant.

Effect of Changes in Production

Harvey Company Year One

Number of units produced	30,000
Number of units sold	25,000
Unit sales price	\$ 30

Variable costs per unit:

Direct materials, direct labor variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3

Fixed costs per year:

Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

Unit Cost Computations for Year One

Unit **product cost** is determined as follows:

	Absorption Costing	Variable Costing
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead (\$150,000 ÷ 30,000 units)	5	-
Unit product cost	\$ 15	\$ 10

Since the number of units produced increased in this example, while the fixed manufacturing overhead remained the same, the absorption unit cost is less.

Absorption Costing: Year One

	Absorption Costing	
Sales (25,000 × \$30)		\$ 750,000
Less cost of goods sold:		
Beginning inventory	\$ -	
Add COGM (30,000 × \$15)	450,000	
Goods available for sale	<u>450,000</u>	
Ending inventory (5,000 × \$15)	<u>75,000</u>	<u>375,000</u>
Gross margin		375,000
Less selling & admin. exp.		
Variable (25,000 × \$3)	\$ 75,000	
Fixed	<u>100,000</u>	<u>175,000</u>
Net operating income		<u><u>\$ 200,000</u></u>

Variable Costing: Year One

		Variable Costing	
		<u>\$ 750,000</u>	
Sales (25,000 × \$30)			
Less variable expenses:			
Beginning inventory	\$ -		
Add COGM (30,000 × \$10)	<u>300,000</u>		
Goods available for sale	300,000		
Less ending inventory (5,000 × \$10)	<u>50,000</u>		
Variable cost of goods sold	250,000		
Variable selling & administrative expenses (25,000 × \$3)	<u>75,000</u>		
Contribution margin		<u>325,000</u>	
Less fixed expenses:			<u>425,000</u>
Manufacturing overhead	\$ 150,000		
Selling & administrative expenses	<u>100,000</u>		<u>250,000</u>
Net operating income			<u><u>\$ 175,000</u></u>

**Variable
manufacturing
costs only.**

**All fixed
manufacturing
overhead is
expensed.**

Effect of Changes in Production

Harvey Company Year Two

Number of units produced	20,000
Number of units sold	25,000
Units in beginning inventory	5,000
Unit sales price	\$ 30

Variable costs per unit:

Direct materials, direct labor variable mfg. overhead	\$ 10
Selling & administrative expenses	\$ 3

Fixed costs per year:

Manufacturing overhead	\$ 150,000
Selling & administrative expenses	\$ 100,000

Unit Cost Computations for Year Two

Unit **product cost** is determined as follows:

	Absorption Costing	Variable Costing
Direct materials, direct labor, and variable mfg. overhead	\$ 10	\$ 10
Fixed mfg. overhead (\$150,000 ÷ 20,000 units)	7.50	-
Unit product cost	\$ 17.50	\$ 10

Since the number of units produced decreased in the second year, while the fixed manufacturing overhead remained the same, the absorption unit cost is now higher.

Absorption Costing: Year Two

	Absorption Costing	
Sales (25,000 × \$30)		\$ 750,000
Less cost of goods sold:		
Beg. inventory (5,000 × \$15)	\$ 75,000	
Add COGM (20,000 × \$17.50)	350,000	
Goods available for sale	425,000	
Less ending inventory	-	425,000
Gross margin		325,000
Less selling & admin. exp.		
Variable (25,000 × \$3)	\$ 75,000	
Fixed	100,000	175,000
Net operating income		\$ 150,000

These are the 20,000 units produced in the current period at the higher unit cost of \$17.50 each.

Variable Costing: Year Two

**Variable
manufacturing
costs only.**

Variable Costing

Sales (25,000 × \$30)		\$ 750,000
Less variable expenses:		
Beg. inventory (5,000 × \$10)	\$ 50,000	
Add COGM (20,000 × \$10)	200,000	
Goods available for sale	<u>250,000</u>	
Less ending inventory	-	
Variable cost of goods sold	<u>250,000</u>	
Variable selling & administrative expenses (25,000 × \$3)	75,000	
Contribution margin		<u>325,000</u>
Less fixed expenses:		425,000
Manufacturing overhead	\$ 150,000	
Selling & administrative expenses	<u>100,000</u>	<u>250,000</u>
Net operating income		<u><u>\$ 175,000</u></u>

**All fixed
manufacturing
overhead is
expensed.**

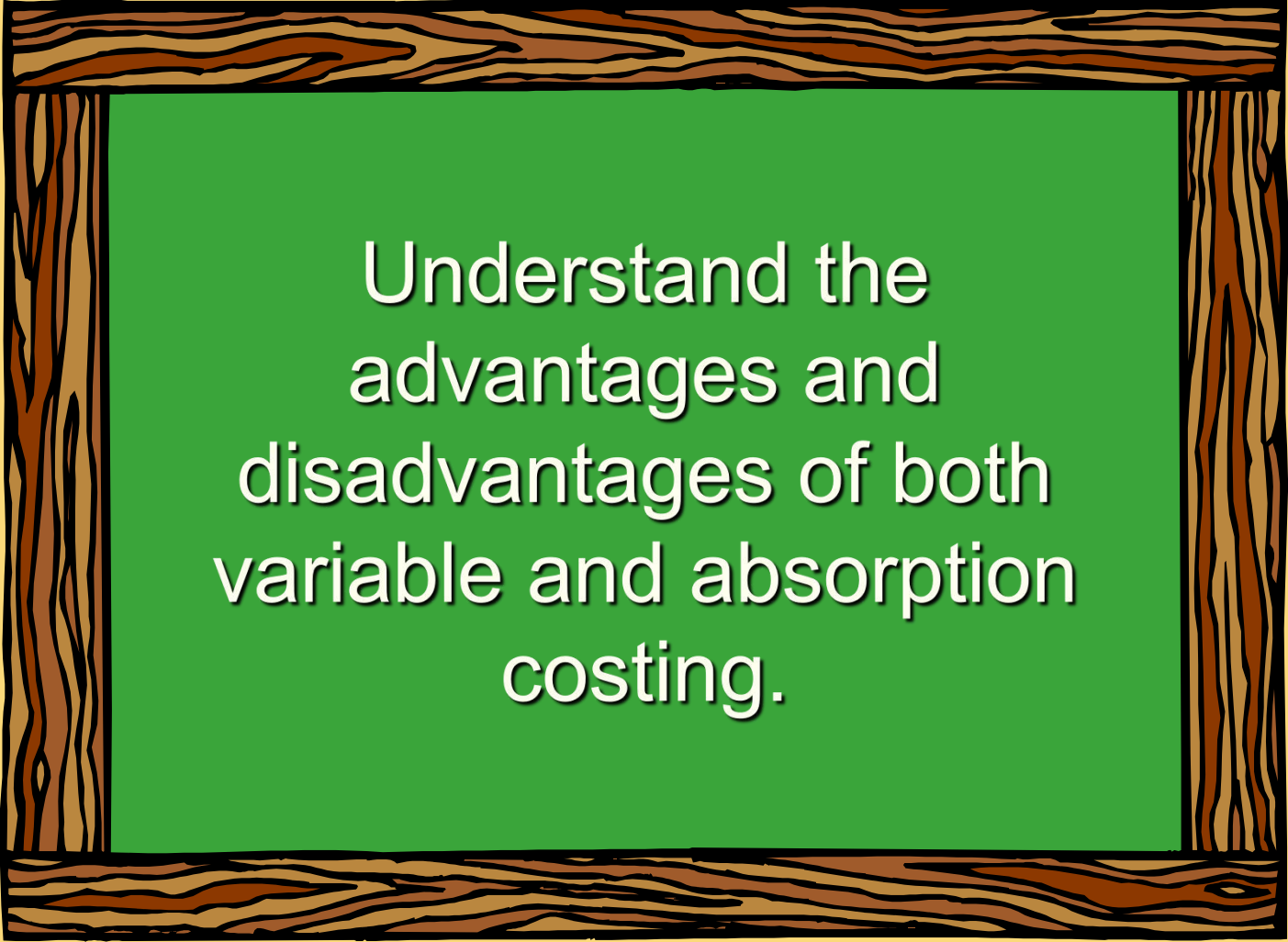
Comparing the Two Methods

Costing Method	Year One	Year Two	Total
Absorption	\$ 200,000	\$ 150,000	\$ 350,000
Variable	175,000	175,000	350,000

Conclusions

- 1 Net operating income **is not affected** by changes in production using variable costing.
- 2 Net operating income **is affected** by changes in production using absorption costing even though the number of units sold is the same each year.

Learning Objective 4

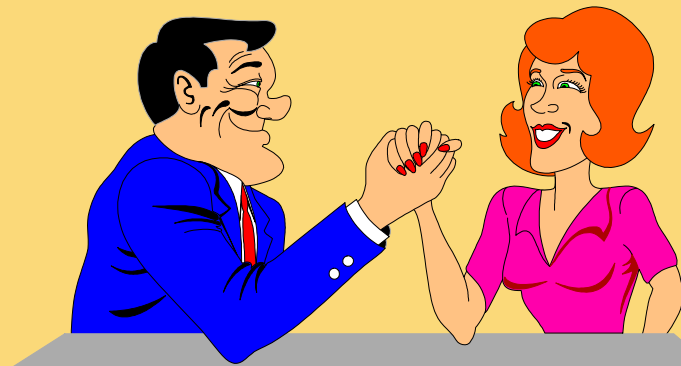


Understand the
advantages and
disadvantages of both
variable and absorption
costing.

Impact on the Manager

Opponents of absorption costing argue that shifting fixed manufacturing overhead costs between periods can lead to faulty decisions.

These opponents argue that variable costing income statements are easier to understand because net operating income is only affected by changes in unit sales. This produces net operating income figures that are more consistent with managers' expectations.



CVP Analysis, Decision Making and Absorption costing

Absorption costing does not support CVP analysis because it essentially treats fixed manufacturing overhead as a variable cost by assigning a per unit amount of the fixed overhead to each unit of production.

Treating fixed manufacturing overhead as a variable cost can:

- Lead to faulty pricing decisions and keep-or-drop decisions.
- Produce positive net operating income even when the number of units sold is less than the breakeven point.

External Reporting and Income Taxes

To conform to GAAP requirements, absorption costing must be used for external financial reports in the United States.

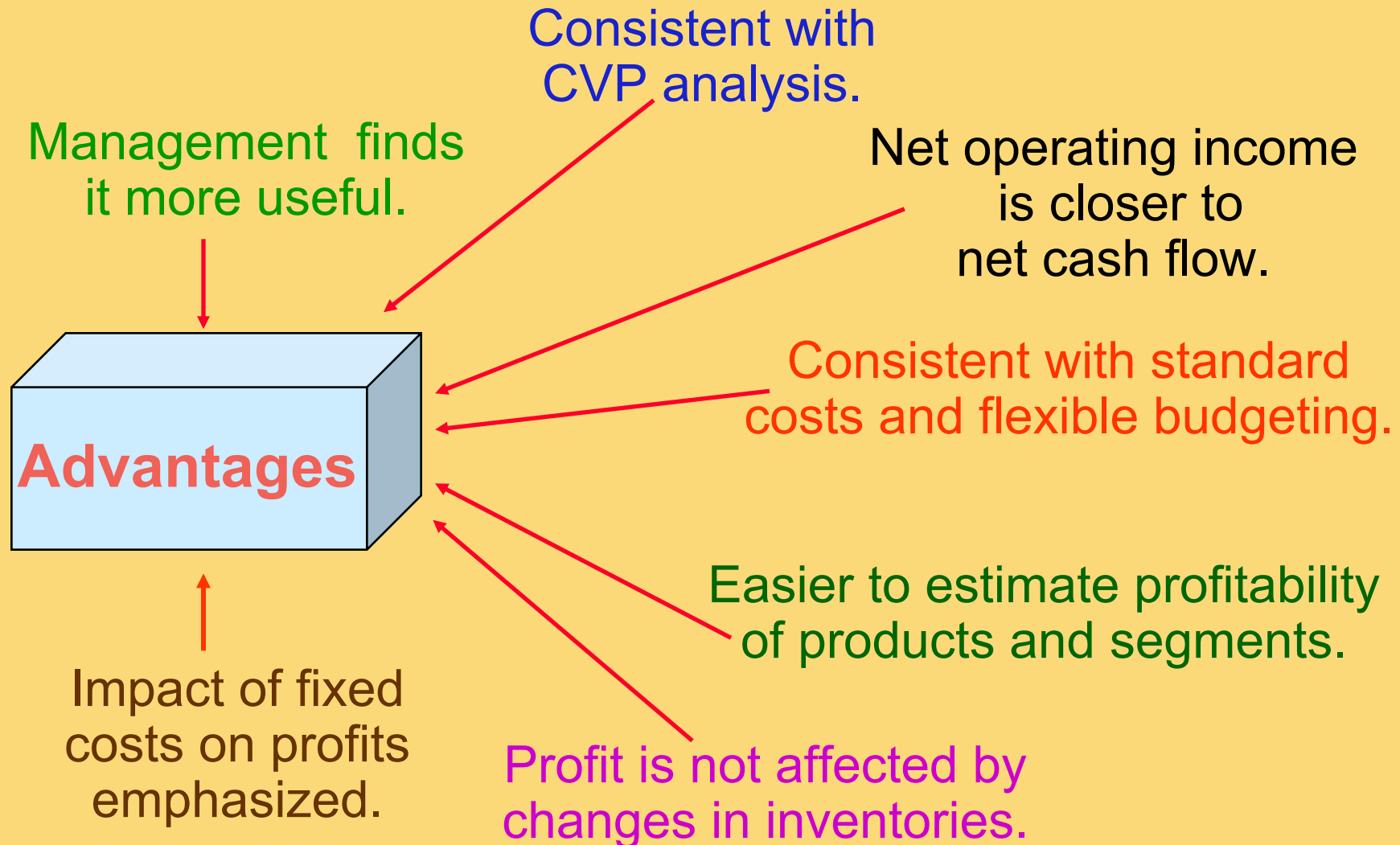


Since top executives are usually evaluated based on external reports to shareholders, they may feel that decisions should be based on absorption cost income.

Under the Tax Reform Act of 1986, absorption costing must be used when filing income tax returns.



Advantages of Variable Costing and the Contribution Approach



Advantages of Variable Costing and the Contribution Approach

- The data required for CVP analysis can be taken directly from a contribution format income statement.
- Profits move in the same direction as sales, assuming other things remain the same.
- Managers often assume that unit product costs are variable. Under variable costing, this assumption is true.
- Fixed costs appear explicitly on a contribution format income statement; thus the impact of fixed costs on profits is emphasized.
- Variable costing data make it easier to estimate the profitability of products, customers, and other business segments.
- Variable costing ties in with cost control methods, such as standard costs and flexible budgeting.
- Variable costing net operating income is closer to net cash flow than absorption costing net operating income.

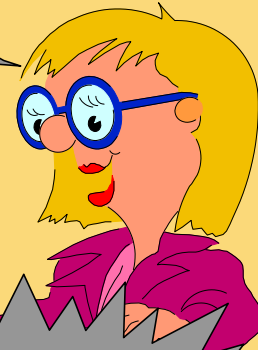
Variable versus Absorption Costing

Fixed manufacturing costs must be assigned to products to properly match revenues and costs.



**Absorption
Costing**

Fixed manufacturing costs are capacity costs and will be incurred even if nothing is produced.



**Variable
Costing**

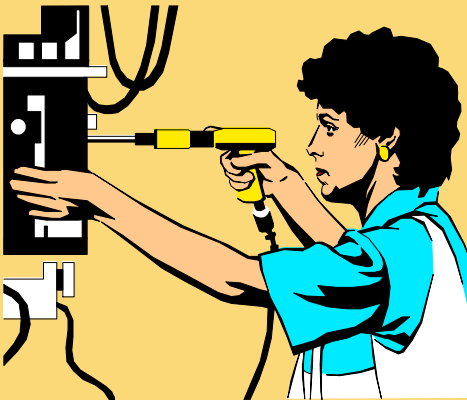
Variable Costing and the Theory of Constraints (TOC)

Companies involved in TOC use a form of variable costing. However, one difference of the TOC approach is that it treats direct labor as a fixed cost for three reasons:

- ❶ Many companies have a commitment to guarantee workers a minimum number of paid hours.
- ❷ Direct labor is usually not the constraint.
- ❸ TOC emphasizes the role direct laborers play in driving continuous improvement. Since layoffs often devastate morale, managers involved in TOC are extremely reluctant to lay off employees.

Impact of JIT Inventory Methods

In a JIT inventory system . . .



**Production
tends to equal
sales . . .**



**So, the difference between variable and
absorption income tends to disappear.**

End of Chapter 7

